

SMART INDIA HACKATHON

Internal Hackathon 2026

GOVERNMENT ENGINEERING COLLEGE, VAISHALI

MASTER PROBLEM STATEMENT CATALOGUE

PS1: AI-Driven Disaster Rescue Resource Allocator

CATEGORY: SOFTWARE | DOMAIN: DISASTER MANAGEMENT

The Problem: During natural disasters, relief materials, medical personnel, and rescue teams are deployed chaotically. There is a lack of real-time, data-driven resource allocation, leading to oversupply in accessible areas and critical shortages in isolated zones.

Expectation from Students: A dynamic web or mobile dashboard that intakes simulated distress data and maps out optimized delivery routes.

The Catch: It must have offline-first capabilities. If their app breaks the second the internet drops, they fail. Expect a working sync-queue architecture, not just a static UI.

PS2: Automated Academic Document Verification via Cryptography

CATEGORY: SOFTWARE | DOMAIN: CYBERSECURITY

The Problem: The proliferation of fake degrees and certificates is a major issue for employers and higher education institutions. Manual verification is incredibly slow and prone to human error.

Expectation from Students: A portal where universities can issue certificates utilizing cryptographic hashing (or a lightweight blockchain) and employers can verify them instantly by uploading a PDF.

The Catch: Expect a working hash-generation and verification flow. No hardcoded 'Verified!' pop-ups allowed.

PS3: Vernacular Financial & Legal Literacy Gamification

CATEGORY: SOFTWARE | DOMAIN: CYBERSECURITY

The Problem: Citizens in rural areas often fall victim to financial fraud or miss out on government schemes due to a lack of accessible, easy-to-understand information in their native languages.

Expectation from Students: A mobile application that teaches basic banking, loan management, and scam prevention through interactive, gamified modules in local languages (e.g., Hindi, Bhojpuri).

The Catch: The architecture must support easy scaling for new languages. Expect a working translation pipeline, not just a few localized strings.

PS4: Dashcam-Based Automated Pothole Mapping

CATEGORY: SOFTWARE | DOMAIN: SMART MOBILITY

The Problem: Municipalities struggle to maintain up-to-date records of road conditions. Manual surveys are inefficient and costly.

Expectation from Students: A computer vision pipeline that processes standard 1080p video (simulating a dashcam feed), detects potholes, and logs their GPS coordinates to a central municipal dashboard.

The Catch: The model must run efficiently. If it takes 10 minutes to process 1 minute of video, it is not viable.

PS5: Smart Contract Agriculture Supply Chain

CATEGORY: SOFTWARE | DOMAIN: AGRICULTURE / AGRITECH

The Problem: Farmers are frequently exploited by middlemen, and consumers have no visibility into the origin of their produce.

Expectation from Students: A decentralized application (dApp) or a highly secure centralized ledger tracking the journey of a crop from farm to table, ensuring transparent pricing and origin tracking.

The Catch: They need to demonstrate how the data is inputted at each stage (farmer, distributor, retailer) seamlessly without requiring the farmer to navigate complex interfaces.

PS6: Predictive Maintenance Dashboard for Campus Infrastructure

CATEGORY: SOFTWARE | DOMAIN: WATER MANAGEMENT

The Problem: Institutional infrastructure (water pumps, transformers, HVAC systems) usually runs until it fails, causing massive disruptions.

Expectation from Students: A software platform that consumes simulated IoT data (temperature, vibration, power draw) from heavy machinery and uses a machine learning model to predict time-to-failure.

The Catch: A real statistical or ML model doing the prediction based on the data stream is expected, not a random number generator triggering a 'maintenance required' alert.

PS7: Dynamic Traffic Signal Optimization via Live Camera Feeds

CATEGORY: SOFTWARE | DOMAIN: SMART MOBILITY

The Problem: Static traffic light timers cause unnecessary congestion, making commuters wait at red lights when crossing lanes are completely empty.

Expectation from Students: An algorithm that analyzes a live video feed of an intersection, counts vehicle density, and adjusts the red/green light durations in real-time to optimize flow.

The Catch: They must successfully handle dynamic edge cases like emergency vehicles.

PS8: Hybrid Green Energy Harvester for Urban Micro-Grids

CATEGORY: HARDWARE | DOMAIN: ENERGY MANAGEMENT

The Problem: Urban environments waste massive amounts of kinetic and ambient energy. There is a need for decentralized, micro-power generation to offset the grid load for municipal infrastructure like streetlights.

Expectation from Students: A functional, physical prototype combining micro-wind turbines (e.g., vertical axis) and solar panels, integrated with a battery management system (BMS).

The Catch: The hardware must actually charge a battery and switch power sources efficiently. Flimsy breadboard connections will not pass evaluation.

PS9: Autonomous Indoor Navigation Rover for Warehousing

CATEGORY: HARDWARE | DOMAIN: AI / NLP / E-GOVERNANCE

The Problem: E-commerce logistics demand high-speed retrieval. Relying solely on human labor or massively expensive imported robotics is inefficient for mid-sized Indian warehouses.

Expectation from Students: A small-scale, autonomous physical rover capable of solving a dynamic maze representing a warehouse floor. It must navigate from Point A to Point B avoiding unexpectedly placed obstacles.

The Catch: True autonomy is required. No remote controllers. It must map or react to its environment using physical sensors (LiDAR, ultrasonic, or camera).

PS10: Low-Cost, IoT-Enabled Water Quality Buoy

CATEGORY: IOT | DOMAIN: WATER MANAGEMENT

The Problem: Continuous monitoring of river and lake pollution is required to combat industrial dumping, but current commercial sensors are prohibitively expensive to deploy en masse.

Expectation from Students: A floating, waterproof hardware device equipped with basic sensors (pH, turbidity, temperature) that continuously transmits data to a remote cloud server.

The Catch: It must survive a splash test. The data transmission must also work seamlessly—hardware execution means nothing if the cloud dashboard remains empty.

PS11: Rural Health Triage Assistant

CATEGORY: SOFTWARE | DOMAIN: HEALTHCARE

The Problem: Rural patients often lack immediate medical guidance, leading to either delayed care for critical issues or overburdening of distant hospitals for minor ailments.

Expectation from Students: A multilingual decision-support system that evaluates user symptoms and determines the urgency of seeking medical care, directing them to appropriate healthcare facilities.

The Catch: The system must explicitly state it is not a doctor. It must demonstrate accurate symptom mapping to a recognized medical triage protocol, avoiding simple keyword matching.

PS12: Fake Job/Internship Detection

CATEGORY: SOFTWARE | DOMAIN: CYBERSECURITY

The Problem: Students are frequently targeted by fraudulent job and internship offers that extract 'processing fees' or steal personal information.

Expectation from Students: A web extension or software portal that analyzes job descriptions, company URLs, and contact details to flag potential scams for students.

The Catch: The detection algorithm must look beyond basic keyword matching; it should ideally analyze domain age, cross-reference blacklists, or use NLP for sentiment/urgency analysis.

PS13: Deepfake + Scam Detection

CATEGORY: SOFTWARE | DOMAIN: CYBERSECURITY

The Problem: Financial scams utilizing AI-generated deepfake voices and manipulated videos are surging, tricking individuals into sending money.

Expectation from Students: A tool combining AI and cybersecurity with multilingual support to detect audio and video manipulation in real-time or via file uploads.

The Catch: Must effectively detect subtle generation artifacts (e.g., spectral audio anomalies, inconsistent lighting) and output a transparent confidence score.

PS14: Regional-Language Misinformation Detector

CATEGORY: SOFTWARE | DOMAIN: CYBERSECURITY

The Problem: Fake news spreads rapidly in Indian regional languages due to a critical gap in AI evaluation models trained for local cultural and linguistic contexts.

Expectation from Students: An AI fact-checking system specifically designed to evaluate misinformation in languages like Hindi, mapping claims against verified sources.

The Catch: Must cross-reference claims against verified regional news APIs or databases, rather than just relying on baseline LLM training weights.

PS15: Smart Heatwave/Disaster Warning System

CATEGORY: SOFTWARE | DOMAIN: DISASTER MANAGEMENT

The Problem: Vulnerable communities lack hyper-localized, actionable early warnings for extreme weather events like heatwaves, leading to preventable casualties.

Expectation from Students: A system combining AI and weather/sensor data to issue map-based, real-time alerts regarding imminent environmental threats.

The Catch: Alerts must provide contextual action plans (e.g., routing to the nearest cooling center), not just serve as a weather API wrapper.

PS16: Smart Classroom Occupancy & Energy Management

CATEGORY: IOT | DOMAIN: ENERGY MANAGEMENT

The Problem: Colleges waste massive amounts of electricity by leaving lights, fans, ACs, and projectors running when classrooms and labs are empty.

Expectation from Students: A hardware-software integration that detects occupancy and environmental conditions to automatically switch off or regulate electrical equipment.

The Catch: Must account for stationary students (e.g., during exams); simple PIR motion sensors that turn off lights when people sit still will be heavily penalized.

PS17: GramTech Village Digital Infrastructure Monitor

CATEGORY: IOT | DOMAIN: WATER MANAGEMENT

The Problem: Rural infrastructure breakdowns (broken pumps, power outages, water contamination) go unnoticed by central authorities for extended periods.

Expectation from Students: Low-cost hardware nodes monitoring water quality, levels, electricity, and weather, feeding a live dashboard for village authorities.

The Catch: Hardware nodes must be heavily power-optimized to run on minimal solar/battery setups for extended periods without manual intervention.

PS18: Offline-First Village Information System

CATEGORY: SOFTWARE | DOMAIN: AI / NLP / E-GOVERNANCE

The Problem: E-governance services and critical information portals fail entirely in rural areas experiencing intermittent or zero internet connectivity.

Expectation from Students: A system providing access to government service information that updates in the background when connected, but functions fully offline.

The Catch: Must implement aggressive caching and a local database architecture. Simple web-views will be instantly rejected.

PS19: Real-Life College RPG (Role-Playing Game)

CATEGORY: SOFTWARE | DOMAIN: SMART CAMPUS / EDTECH

The Problem: Student engagement in productive extracurriculars, projects, and volunteering is low due to a lack of immediate reward or tangible tracking.

Expectation from Students: A gamified progression system converting college activities (coding, clubs, events) into quests, XP, and unlockable achievements.

The Catch: Needs a rigorous, verifiable proof-of-work mechanism for quests, ensuring students cannot arbitrarily claim completion for free XP.

PS20: Smart Campus Complaint Management System

CATEGORY: SOFTWARE | DOMAIN: WATER MANAGEMENT

The Problem: Campus infrastructure issues (hostel maintenance, classroom electricity, water) are reported via untracked manual registers, leading to unresolved grievances.

Expectation from Students: A unified platform for students and staff to submit complaints, track their resolution, and handle automated escalations.

The Catch: Must include a strict Role-Based Access Control (RBAC) system and automated escalation matrices if Service Level Agreements (SLAs) are breached.

PS21: Integrated Smart Academic Management Platform

CATEGORY: SOFTWARE | DOMAIN: SMART CAMPUS / EDTECH

The Problem: Academic management, attendance tracking, and student performance metrics are scattered across multiple incompatible, legacy software systems.

Expectation from Students: A cohesive platform consolidating academic management and student performance data into actionable dashboards for faculty and students.

The Catch: Must demonstrate API readiness to integrate with existing legacy databases safely, rather than functioning as a closed-off, standalone silo.

PS22: Infrastructure Digital Twin

CATEGORY: SOFTWARE | DOMAIN: SMART CAMPUS / EDTECH

The Problem: Facility managers lack a real-time, visual, and centralized overview of campus infrastructure, spatial utilization, and equipment status.

Expectation from Students: A digital representation (2D/3D map or dashboard) tracking rooms, equipment, and facilities dynamically.

The Catch: State changes in the real world (e.g., a specific lab's AC failing) must reflect accurately and in real-time on the digital twin.

PS23: Panchayat Development Monitoring System

CATEGORY: SOFTWARE | DOMAIN: SMART MOBILITY

The Problem: Villagers have no transparency regarding the progress of funded local projects (roads, drains, streetlights), leading to poor accountability.

Expectation from Students: A digital dashboard tracking village project lifecycles, allowing citizens to report problems using geo-tagged photographs.

The Catch: Geo-tagging and timestamping must be spoof-proof to prevent contractors from uploading fake, manipulated, or recycled images.

PS24: Rural Waste Collection Optimizer

CATEGORY: SOFTWARE | DOMAIN: AI / NLP / E-GOVERNANCE

The Problem: Panchayat-level waste management is ad-hoc, leading to overflowing garbage hotspots, missed collections, and wasted fuel for transport teams.

Expectation from Students: A mapping system that identifies garbage hotspots and calculates the most optimized, fuel-efficient collection routes.

The Catch: Must dynamically handle the Traveling Salesperson Problem efficiently, allowing for on-the-fly recalculations as new hotspots are reported.

PS25: Electricity Wastage Anomaly Detector

CATEGORY: SOFTWARE | DOMAIN: ENERGY MANAGEMENT

The Problem: Identifying which specific appliances or sectors are wasting power in a large institutional grid is nearly impossible with standard metering.

Expectation from Students: A system monitoring consumption that detects abnormal, unnecessary usage patterns and sends alerts or triggers automatic shut-offs.

The Catch: Must utilize disaggregation techniques to identify individual high-draw appliances from a single smart meter data stream.

PS26: Flood/Disaster Offline Rescue Network

CATEGORY: IOT | DOMAIN: DISASTER MANAGEMENT

The Problem: Cellular and internet networks frequently fail entirely during major disasters, paralyzing rescue operations and stranding victims.

Expectation from Students: A local mesh or LoRa network protocol connecting rescue teams and affected people without relying on traditional ISP infrastructure.

The Catch: Must physically prove packet transmission (text or telemetry) between at least three distinct nodes without any active Wi-Fi or LTE connection.

PS27: Women Safety Analytics

CATEGORY: SOFTWARE | DOMAIN: AI / NLP / E-GOVERNANCE

The Problem: Existing safety applications require manual SOS triggering, which is often impossible during an active assault or immediate threat.

Expectation from Students: A system that automatically detects unsafe situations, triggers real-time alerts, coordinates emergency responses, and maps unsafe areas.

The Catch: The automated trigger (e.g., via acoustic scream detection or anomalous movement) must maintain an exceptionally low false-positive rate.

PS28: Offline Emergency Communication Protocol

CATEGORY: SOFTWARE | DOMAIN: AI / NLP / E-GOVERNANCE

The Problem: Victims trapped in areas without network coverage cannot send SOS signals, messages, or their GPS coordinates to nearby helpers.

Expectation from Students: A mobile application utilizing peer-to-peer technologies to hop SOS messages and locations across nearby devices without internet.

The Catch: Must demonstrate multi-hop message relay capabilities. A simple point-to-point single connection is insufficient for evaluation.

PS29: People Trapped in Collapsed Buildings Detector

CATEGORY: HARDWARE | DOMAIN: DISASTER MANAGEMENT

The Problem: Rescuers waste critical 'golden hours' clearing empty debris because they lack systems to pinpoint exact human locations underneath.

Expectation from Students: A hardware/software solution detecting human presence under debris and providing exact or approximate coordinates to rescuers.

The Catch: The sensor logic (acoustic, thermal, or radio) must actively differentiate between human life signs and environmental/animal noise.

PS30: Railway Track Defect Monitor

CATEGORY: IOT | DOMAIN: AI / NLP / E-GOVERNANCE

The Problem: Manual railway track inspections are too slow and infrequent to prevent sudden derailments caused by micro-cracks or deformations.

Expectation from Students: An automated monitoring system (drone or train-mounted) continuously detecting cracks and abnormal vibrations, issuing early alerts.

The Catch: If simulated, the anomaly detection model must prove it can handle the extreme visual and vibrational noise typical of moving trains.

PS31: Transformer Failure Prediction IoT

CATEGORY: IOT | DOMAIN: AI / NLP / E-GOVERNANCE

The Problem: Grid transformers fail abruptly due to overload and wear, causing massive regional blackouts and incurring high replacement costs.

Expectation from Students: An IoT-based monitor checking transformer conditions (temperature, vibration) to predict failure and schedule preventative maintenance.

The Catch: The dashboard must calculate and display a clear 'Remaining Useful Life (RUL)' metric based on historical degradation models, not just raw sensor data.

PS32: Flood Early Warning for Villages

CATEGORY: SOFTWARE | DOMAIN: DISASTER MANAGEMENT

The Problem: Sudden local flooding catches rural communities off guard due to a lack of hyper-local water level and rainfall predictive modeling.

Expectation from Students: A system monitoring water levels and rainfall to predict local flood risks and provide advance warnings to villagers.

The Catch: The predictive model must factor in local topography and drainage capacity, not just generic rainfall volume.

PS33: Deepfake Harassment Evidence Preserver

CATEGORY: SOFTWARE | DOMAIN: CYBERSECURITY

The Problem: Victims of deepfake harassment struggle to safely log, verify, and preserve manipulated media as legal evidence before attackers delete it.

Expectation from Students: A secure portal for victims to report manipulated media, which verifies tampering and preserves evidence for legal use.

The Catch: Chain of custody must be preserved via cryptographic hashing; robust metadata extraction and immutable logging are mandatory.

PS34: Government Scheme Eligibility Matcher

CATEGORY: SOFTWARE | DOMAIN: AI / NLP / E-GOVERNANCE

The Problem: Citizens struggle to navigate complex bureaucratic criteria and disparate portals to find out which government schemes they qualify for.

Expectation from Students: An automated system taking basic citizen details to identify relevant schemes, outlining eligibility, required documents, and application processes.

The Catch: The underlying rules engine must be designed so that non-programmers (administrators) can easily update criteria when government policies change.

PS35: Campus Carbon Footprint Calculator

CATEGORY: SOFTWARE | DOMAIN: SMART CAMPUS / EDTECH

The Problem: Educational institutions have no standard, automated way to measure or optimize their daily carbon emissions from various operational sectors.

Expectation from Students: A calculator estimating campus carbon emissions based on precise inputs for electricity, transport, waste management, and infrastructure.

The Catch: The calculation logic must strictly adhere to established environmental frameworks (e.g., GHG Protocol), rather than using arbitrary math.

PS36: AI-Based Crop Disease Detection & Treatment

CATEGORY: SOFTWARE | DOMAIN: HEALTHCARE

The Problem: Small farmers lose significant yields because they cannot accurately identify crop diseases early or access expert agricultural advice in time.

Expectation from Students: An application allowing farmers to photograph diseased crops, identifying the disease via AI, and suggesting treatments in simple language (with Hindi voice support).

The Catch: The on-device machine learning model must be lightweight enough to run inferences on budget, older Android smartphones without lagging or requiring cloud compute.

PS37: Smart Water Monitoring Network

CATEGORY: IOT | DOMAIN: WATER MANAGEMENT

The Problem: Institutional and municipal overhead tanks frequently overflow or leak, wasting thousands of liters of clean water daily.

Expectation from Students: A sensor network monitoring tank levels, tracking consumption, detecting pipe leakage through abnormal flow, and automating pump control.

The Catch: Must reliably detect the difference between normal heavy usage (e.g., morning hostel hours) and actual pipe leakage.

PS38: Smart Timetable Generator

CATEGORY: SOFTWARE | DOMAIN: SMART CAMPUS / EDTECH

The Problem: Manual academic timetable creation leads to faculty double-booking, classroom resource clashes, and highly inefficient gaps for students.

Expectation from Students: An algorithm automatically generating conflict-free timetables considering constraints like classrooms, faculty availability, subjects, and student batches.

The Catch: Must mathematically solve for complex edge cases (e.g., combined lectures, consecutive lab constraints) and generate the schedule rapidly.

PS39: AI-Based Fake/Manipulated Document Detection

CATEGORY: SOFTWARE | DOMAIN: CYBERSECURITY

The Problem: Digitally edited certificates, marksheets, and official documents are increasingly difficult to verify manually, facilitating mass fraud.

Expectation from Students: A system analyzing uploaded documents using OCR, layout consistency, metadata, and image forensics to produce a tampering-risk score.

The Catch: The forensics pipeline must detect advanced manipulation techniques like localized pixel cloning and font-kerning anomalies, not just missing metadata.

PS40: Local Language Government Service Assistant

CATEGORY: SOFTWARE | DOMAIN: AI / NLP / E-GOVERNANCE

The Problem: Critical information regarding government schemes is often buried in complex English or dense official language PDFs, inaccessible to common citizens.

Expectation from Students: An AI-powered chatbot explaining government schemes and services fluently in Hindi and local regional languages.

The Catch: Must strictly constrain its responses to verified government data to prevent 'hallucinating' non-existent scheme benefits or incorrect deadlines.

PS41: Digital Laboratory Equipment Management

CATEGORY: IOT | DOMAIN: AI / NLP / E-GOVERNANCE

The Problem: Crucial laboratory equipment goes missing, breaks down without notice, or overlaps in usage schedules, heavily disrupting practical sessions.

Expectation from Students: A tracking system managing laboratory equipment availability, logging maintenance schedules, and tracking damaged/missing items.

The Catch: Must include a rapid hardware-integrated check-out/check-in mechanism (e.g., RFID or barcode scanning) rather than relying on manual text entry.

PS42: Autonomous Search-and-Rescue Drone System

CATEGORY: HARDWARE | DOMAIN: DISASTER MANAGEMENT

The Problem: Human rescuers cannot safely navigate or scan heavily flooded, toxic, or collapsed areas fast enough to locate stranded individuals.

Expectation from Students: A drone system utilizing thermal imaging, GPS, and onboard image processing to rapidly locate humans in disaster zones.

The Catch: If physical drone hardware is unfeasible for the venue, a high-fidelity simulation environment (e.g., ROS/Gazebo) proving the vision pipeline is strictly required.

PS43: Flood Evacuation Route Optimizer

CATEGORY: SOFTWARE | DOMAIN: DISASTER MANAGEMENT

The Problem: Standard GPS navigation maps routinely route evacuating users into flooded roads or traffic bottlenecks during disasters.

Expectation from Students: A GIS-based platform identifying safe routes, shelters, hospitals, and transport options explicitly utilizing dynamic flood data.

The Catch: Must ingest simulated 'road block/flood' data and dynamically recalculate paths instantly, completely overriding standard 'fastest' routes if deemed unsafe.

PS44: Cyber Threat Detection for Small Institutions

CATEGORY: SOFTWARE | DOMAIN: CYBERSECURITY

The Problem: Colleges, schools, and small offices cannot afford enterprise-grade firewalls, leaving their networks vulnerable to ransomware and phishing.

Expectation from Students: An affordable, lightweight security monitoring platform detecting malware, unauthorized access, and suspicious network activity.

The Catch: It must demonstrate the ability to run passively on the network (packet sniffing) without crippling the institution's limited bandwidth.

PS45: Mental-Health Screening and Counselling Referral

CATEGORY: SOFTWARE | DOMAIN: HEALTHCARE

The Problem: Students and citizens face severe social stigma and a lack of accessible platforms for preliminary mental health screening.

Expectation from Students: An anonymous platform conducting preliminary psychological screening and securely connecting users with authorized counsellors or helplines.

The Catch: Strict data privacy is paramount; the architecture must prove that no raw, identifiable user screening data is stored in plain text.

PS46: Emergency Medical Resource Allocation

CATEGORY: SOFTWARE | DOMAIN: HEALTHCARE

The Problem: During mass casualty emergencies, finding available hospital beds, matching blood groups, and oxygen supplies is a fragmented, chaotic process.

Expectation from Students: A live, centralized platform mapping available ambulances, hospital beds, blood groups, oxygen supplies, and emergency facilities.

The Catch: It requires a functional, simulated live-updating backend capable of handling high-concurrency requests. Static databases will fail evaluation.

PS47: AI-Based Medical Image Pre-Screening

CATEGORY: SOFTWARE | DOMAIN: HEALTHCARE

The Problem: A severe shortage of radiologists in rural clinics severely delays the preliminary diagnosis of highly treatable common conditions.

Expectation from Students: A decision-support tool for preliminary detection of conditions from X-rays, skin images, or ultrasounds, strictly assisting (not replacing) doctors.

The Catch: Model accuracy metrics (Precision/Recall) must be clearly visualized, and the system must handle negative edge cases gracefully without returning false positives.

PS48: Rural Telemedicine and Diagnostic Support Platform

CATEGORY: SOFTWARE | DOMAIN: AI / NLP / E-GOVERNANCE

The Problem: Remote villages lack continuity of care; patients struggle to maintain digital prescriptions, diagnostic reports, and follow-up tracking with distant doctors.

Expectation from Students: A platform connecting patients, local health workers, and remote doctors, supporting end-to-end digital prescriptions and follow-up care.

The Catch: The system architecture must gracefully support extremely low-bandwidth network environments, gracefully degrading video calls to audio or text.

PS49: Solar-Powered Smart Irrigation Network

CATEGORY: IOT | DOMAIN: AGRICULTURE / AGRITECH

The Problem: Blind, manual irrigation wastes massive amounts of groundwater and damages crops due to over-watering or ignoring impending weather.

Expectation from Students: A system combining soil moisture sensors, weather forecasts, and solar energy to automate and optimize water pumps.

The Catch: Must execute predictive logic (e.g., halting scheduled watering if heavy rain is forecast in 2 hours) rather than relying on simple dry-soil threshold triggers.

PS50: Drone-Based Crop Health Monitoring

CATEGORY: HARDWARE | DOMAIN: HEALTHCARE

The Problem: Manual inspection of large agricultural acreage for early signs of pests or nutrient deficiency is physically impossible for a single farmer.

Expectation from Students: An image-analysis system utilizing drone feeds to rapidly detect water stress, pest attacks, nutrient deficiency, and damaged crop areas.

The Catch: The computer vision pipeline must actively segment the specific disease/stress areas within the image and output an actionable heatmap for the farmer.

PS51: AI-Based Flood and Water-Level Forecasting

CATEGORY: SOFTWARE | DOMAIN: DISASTER MANAGEMENT

The Problem: Traditional flood warnings lack hyper-local predictive accuracy, causing communities to receive alerts either too late or unnecessarily.

Expectation from Students: A prediction model synthesizing rainfall, river-level, drainage, and geographic data to forecast local flooding and alert vulnerable communities.

The Catch: The model's algorithm must incorporate historical drainage performance data, proving it understands that the same rainfall affects different topographies differently.

PS52: Intelligent Document Processing for Government Offices

CATEGORY: SOFTWARE | DOMAIN: AI / NLP / E-GOVERNANCE

The Problem: Government offices are suffocating under physical Hindi and English forms, leading to severe manual processing backlogs and data entry errors.

Expectation from Students: An OCR and NLP platform extracting information from mixed-language forms, classifying the documents, and automatically flagging missing fields.

The Catch: Must demonstrate high accuracy extraction on handwritten Hindi text (Devanagari script), which is notoriously difficult for standard OCR engines.

PS53: AI-Based Missing Person Identification System

CATEGORY: SOFTWARE | DOMAIN: AI / NLP / E-GOVERNANCE

The Problem: Coordinating missing person reports across multiple disconnected shelter, hospital, and police databases is highly inefficient and slow.

Expectation from Students: A privacy-conscious platform matching authorized missing-person records against incoming images from shelters, hospitals, and police databases.

The Catch: Strict facial recognition privacy controls must be embedded; the architecture must instantly purge non-matching images from its processing cache.

PS54: Smart Tourism Safety & Emergency Assistance

CATEGORY: SOFTWARE | DOMAIN: AI / NLP / E-GOVERNANCE

The Problem: Tourists frequently face language barriers, getting lost, or encountering emergencies in unfamiliar regions without access to local knowledge.

Expectation from Students: A platform providing tourists with automated safety alerts, location-based cultural information, and one-tap emergency assistance routing.

The Catch: Must utilize dynamic geofencing to silently push severe alerts only when tourists enter specific, verified high-risk or disaster-prone zones.

PS55: Digital Heritage & Monument Preservation

CATEGORY: SOFTWARE | DOMAIN: AI / NLP / E-GOVERNANCE

The Problem: Historical monuments degrade over time unnoticed, lacking structured digital documentation required for accurate restoration by authorities.

Expectation from Students: A platform to digitally document monument conditions, allowing crowdsourced or official tracking of structural preservation issues.

The Catch: Needs a built-in image-comparison tool to highlight and calculate the percentage of structural degradation over a specific timeline.